



5 – Proofs

Week 6 – 16 Sep 2025

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Cheat sheet

- Direct Proofs ($p \rightarrow q$)
- Indirect Proofs
 - Contraposition ($\neg q \rightarrow \neg p$)
 - Contradiction $\neg(p \rightarrow q)$

Ex. 3.4.4. Use universal instantiation or universal modus ponens to fill in a valid conclusion for the following argument.

$\forall$ real number r , a , and b , if r is positive, then $(r^a)^b = r^{ab}$.

$r = 3$, $a = 1/2$, and $b = 6$ are particular real numbers such that r is positive.

$\therefore$ _____

SOLUTION . $\therefore \left(3^{1/2}\right)^6 = 3^{(\frac{1}{2})(6)}$.

□

Ex. 3.4.17. The argument below could be valid by universal modus ponens or universal modus tollens, or invalid and exhibits the converse or the inverse error. State whether it is valid or invalid. Justify your answer.

If an infinite series converges, then the terms go to 0.

The terms of the infinite series $\sum_{n=1}^{\infty} \frac{1}{n}$ go to 0.

$\therefore$ the infinite series $\sum_{n=1}^{\infty} \frac{1}{n}$ converges.

SOLUTION . Invalid. This argument exhibits the converse error.

□

MH1101 Calculus 2: p-series

***p*-series test** [\[edit\]](#)

A commonly used corollary of the integral test is the p-series test. Let $k > 0$. Then $\sum_{n=k}^{\infty} \left(\frac{1}{n^p} \right)$ converges if $p > 1$.

- For $p = 1$, it's called harmonic series.

Ex. 4.1.13. Disprove the statement by giving a counterexample.

For all integers m and n , if $2m + n$ is odd then m and n are both odd.

SOLUTION . Let $m = 2$ and $n = 1$. Then $2m + n = 2 \cdot 2 + 1 = 5$, which is odd. But m is not odd, and so it is false that both m and n are odd. $\square$

Ex. 4.1.40. Find the mistake(s) in the “proof” shown below.

Theorem: For all integers k , if $k > 0$ then $k^2 + 2k + 1$ is composite.

“Proof: Suppose k is any integer such that $k > 0$. If $k^2 + 2k + 1$ is composite, then $k^2 + 2k + 1 = r \cdot s$ for some integers r and s such that

$$1 < r < (k^2 + 2k + 1)$$

and

$$1 < s < (k^2 + 2k + 1).$$

Since

$$k^2 + 2k + 1 = r \cdot s$$

Wool sweaters are better than nylon
jackets as fall attire

wool sweaters have higher wool content

and both r and s are strictly between 1 and $k^2 + 2k + 1$, then $k^2 + 2k + 1$ is not prime. Hence $k^2 + 2k + 1$ is composite as was to be shown.”

SOLUTION . The incorrect proof is begging the question. There is no justification to why the integers r and s exists, or rather, the justification is assuming (instead of proving) that $k^2 + 2k + 1$ is composite. □

Ex. 4.1.56. Determine whether the statement is true or false. Justify your answer with a proof or a counterexample, as appropriate.

(Two integers are **consecutive** if and only if one is one more than the other.) Any product of four consecutive integers is one less than a perfect square.

$$\text{WTS: } n(n+1)(n+2)(n+3) = M^2 - 1$$

Step 1: Use calculator to verify if this statement is correct

Let's try:

$$100 \cdot 101 \cdot 102 \cdot 103 = 106110600$$

$$\text{Sqrt}(106110600 + 1) = 10301$$

Try a **few numbers** to see if it's correct or not.

Ex. 4.2.8. The zero property says that if a product of two real numbers is 0, then one of the numbers must be 0.

- a. Write this property formally using quantifiers and variables.
- b. Write the contrapositive of your answer to part (a).
- c. Write an informal version (without quantifier symbols or variables) for your answer to part (b).

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SOLUTION . a. $\forall$ real numbers x and y , if $xy = 0$ then $x = 0$ or $y = 0$.

b. $\forall$ real numbers x and y , if $x \neq 0$ and $y \neq 0$ then $xy \neq 0$.

c. If two real numbers are both nonzero, then the product is nonzero.



Definition: A number c is called a **root** of a polynomial $p(x)$ if and only if $p(c) = 0$.

Ex. 4.2.32. Prove that for all real numbers c , if c is a root of a polynomial with rational coefficients, then c is a root of a polynomial with integer coefficients.

The previous question 4.2.31 is skipped

Bonus Qn: Which proving method is the easiest?

Suppose p is prime,
If $p \mid n^2$, then $p \mid n$.